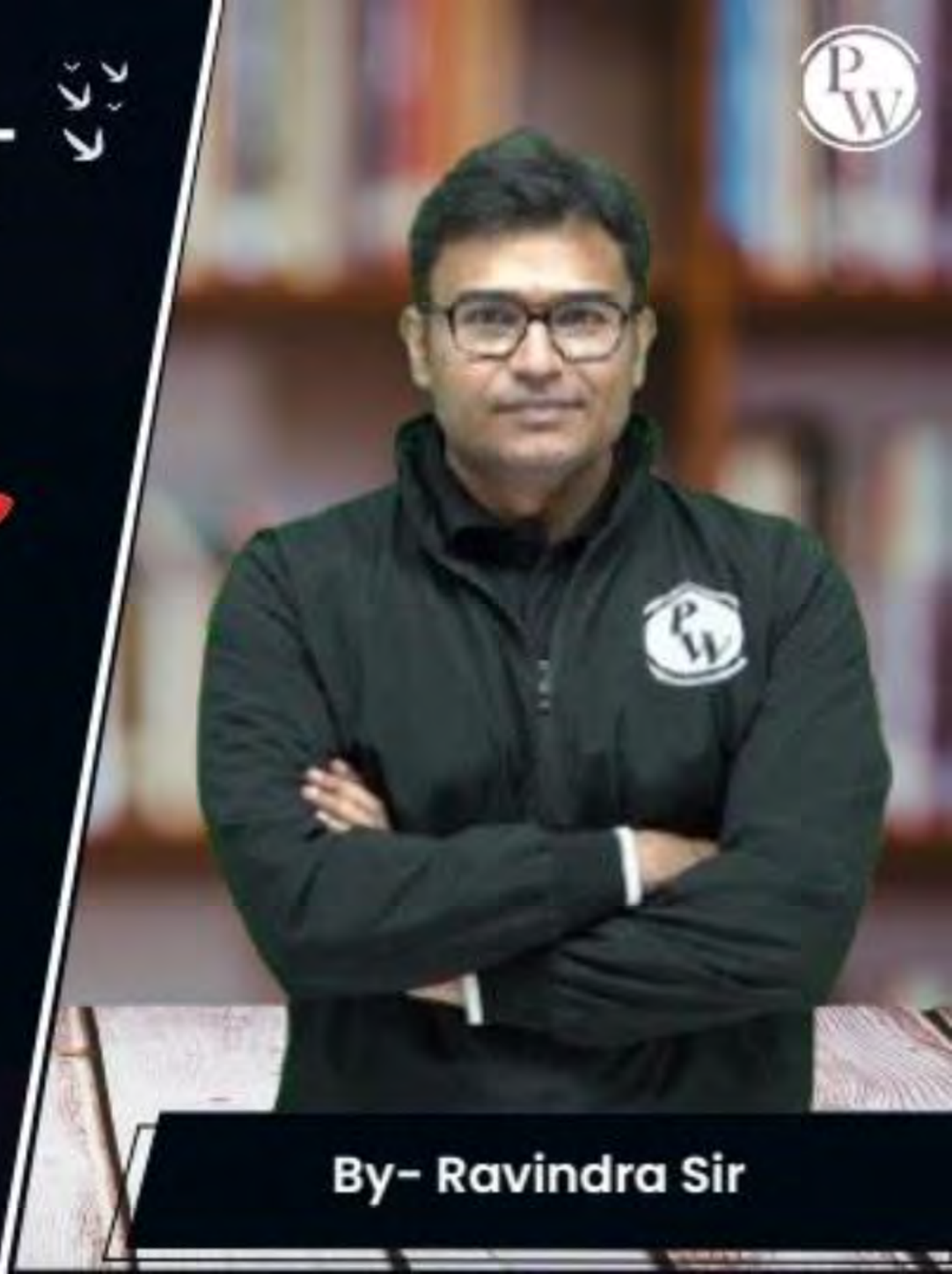
Computer Science & IT

Theory of Computation



CFL

Lecture No. 8



By- Ravindra Sir

Recap of Previous Lecture



- Topic
- Topic
- Topic

CFG, string matching, derivation

✓ $WxcW^R$

Topics to be Covered

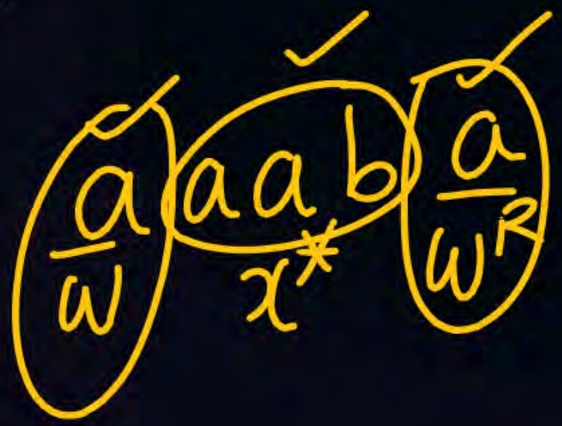


- Topic
- Topic
- Topic

Ambiguity

Ambiguity of grammars :-

③ Rules → may cause ambiguity



Rule ①:- If there are two productions

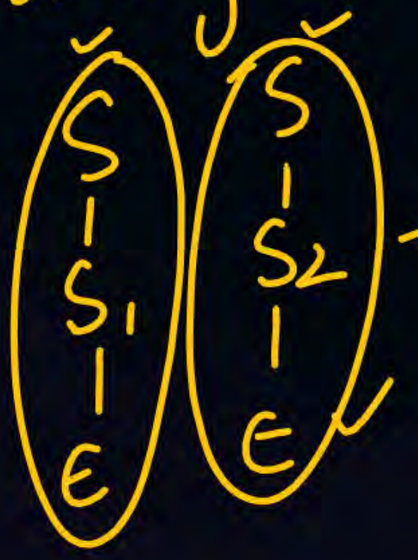
$S \to S_1 / S_2$ and $L(S_1) \cap L(S_2) \neq \phi$

then it is ambiguous

$S \to S_1 / S_2$

$S_1 \to aS_1 / \epsilon$

$S_2 \to bS_2 / \epsilon$



→ ambiguous grammar

$a^*(bb)^* \cap c^*(dd)^* = \{\epsilon\}$

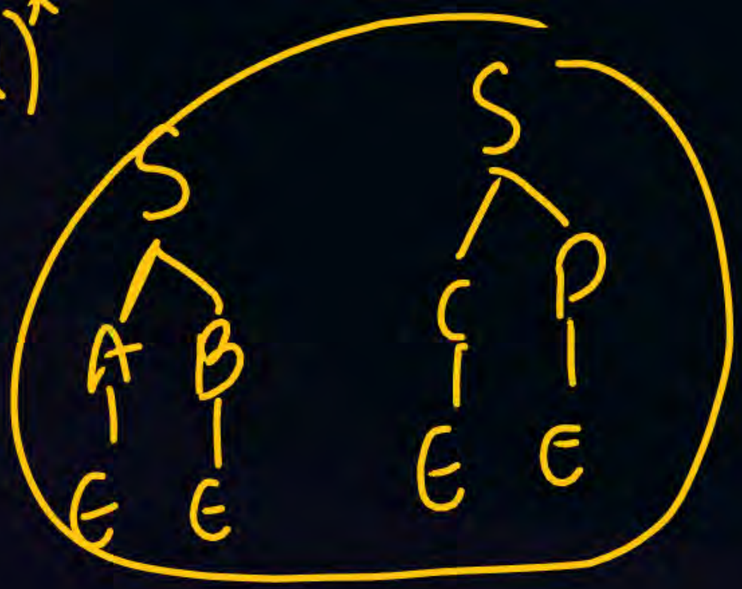
$S \rightarrow AB / CD$

$A \rightarrow aA/\epsilon \Rightarrow A \Rightarrow a^*$

$B \rightarrow bbB/\epsilon \Rightarrow (bb)^*$

$C \rightarrow cC/\epsilon \Rightarrow C^*$

$D \rightarrow ddD/\epsilon \Rightarrow (dd)^*$



ambiguous grammar

$a^*(bb)^* \cap c^+(dd)^* \Rightarrow \phi$

$S \to AB/CD$

$A \to aA/\epsilon \quad A \Rightarrow a^*$

$B \Rightarrow bbB/\epsilon \quad B \Rightarrow (bb)^*$

$C \to cC/c \quad C \Rightarrow c^+$

$D \to ddD/\epsilon \quad d \to (dd)^*$

unamb grammar

Rule ② :-

Left recursion and left factoring may cause ambiguity

not always

$A \rightarrow A\alpha / \beta$

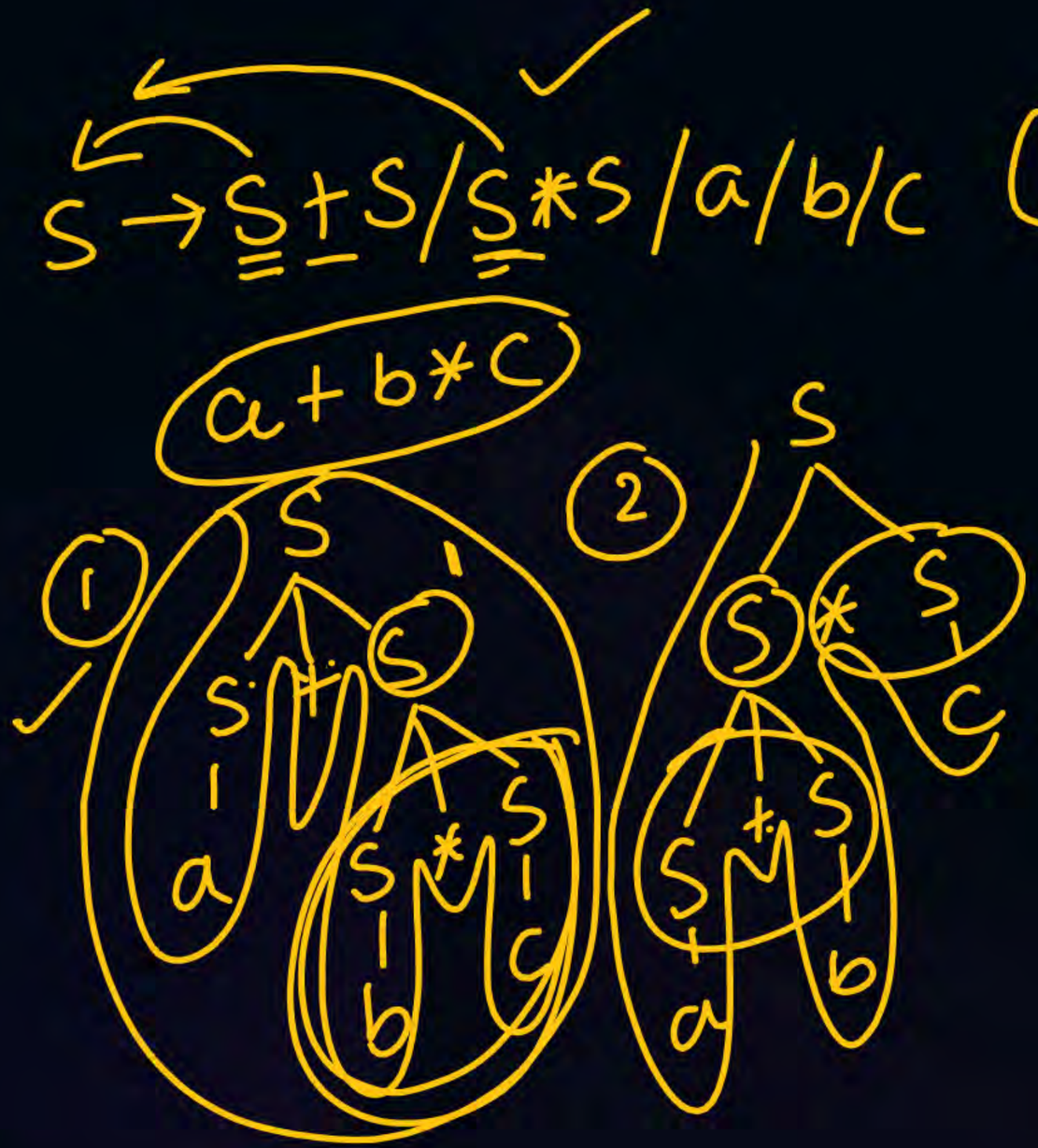
left recursion

$A \rightarrow Aa / \epsilon$

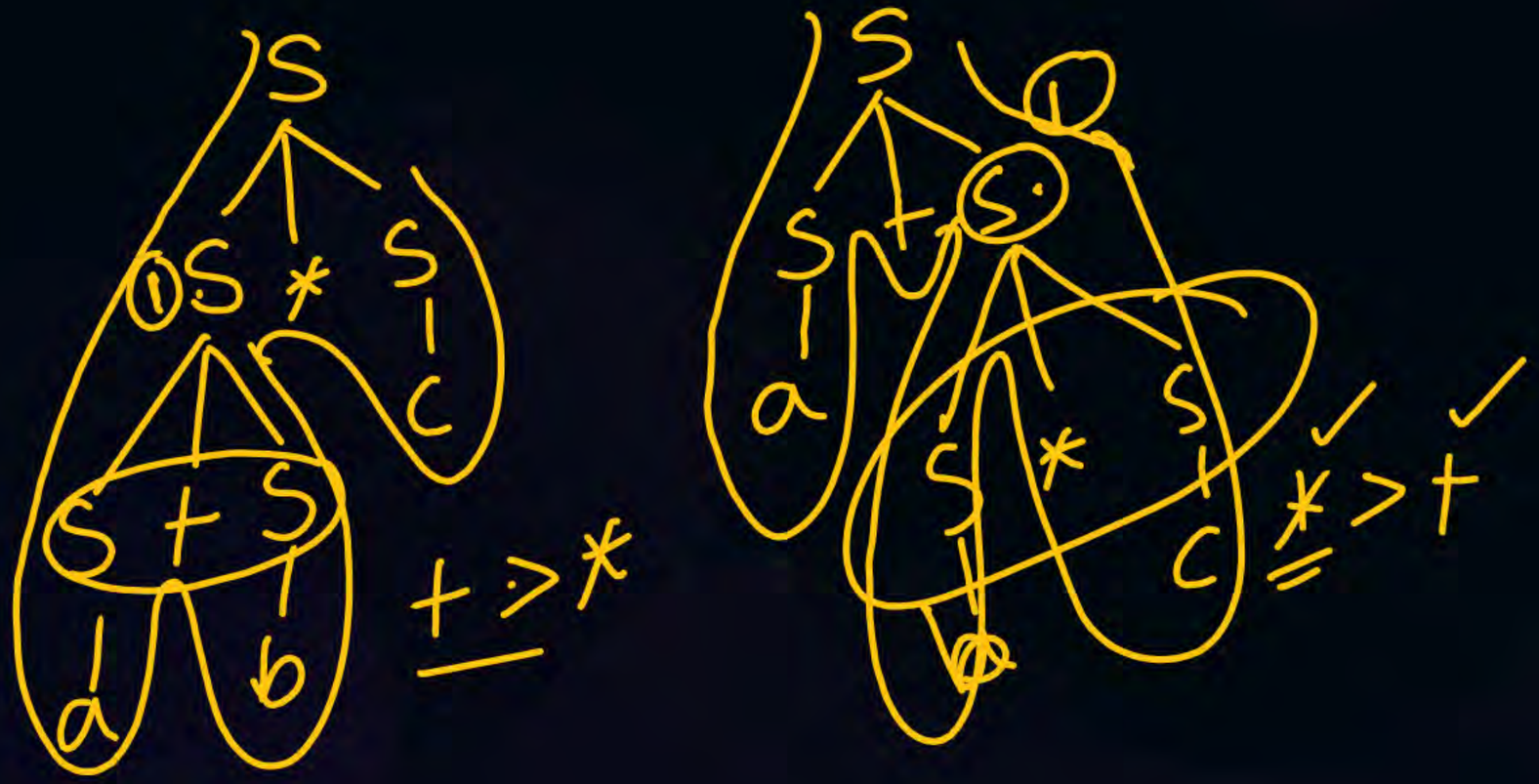
unamb

$S \to S + S / S * S / a / b / c$ (left recursive)

$a + (b * c)$



ambiguous

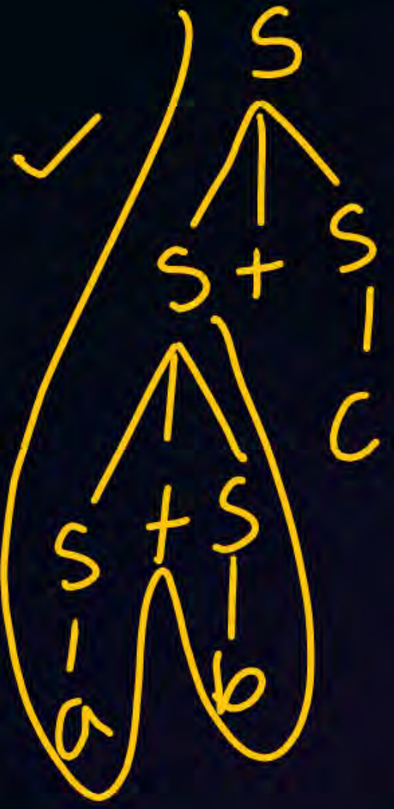

S
S * S
S + S
a b
+ > *
S
S + S
a
S * S
S + S
c
* > +


S 7
1 S + S 6
1
S * S
1 2 1 3
S 9
3 S * S 1 3
S + S
1 1 2

LR

amb $S \rightarrow S+S \mid a \mid b \mid c$

$+ > +$





$S \rightarrow Sa/\epsilon$

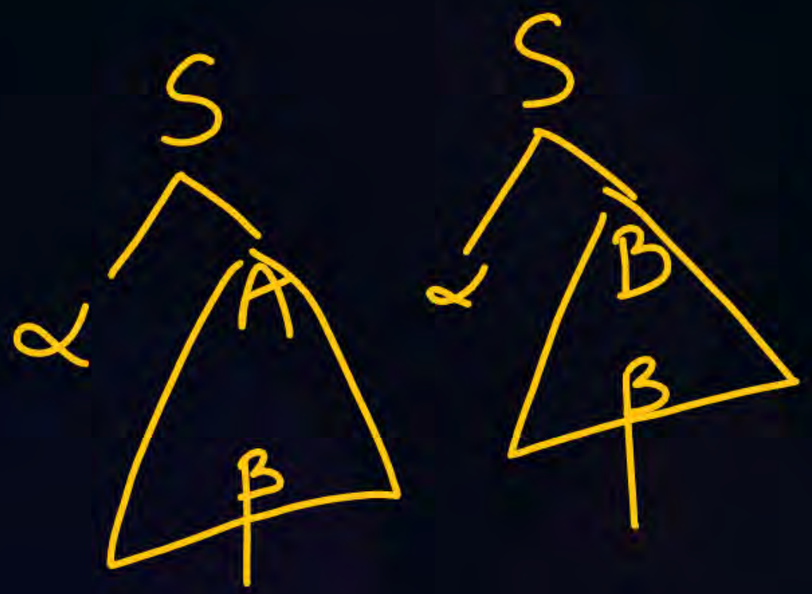
Left recursion may not be ambiguous in all cases

Left factoring :-

$$S \rightarrow ABC / ABD$$

$$S \rightarrow AB / A\,C$$

$S \rightarrow \alpha \tilde{A} \mid \alpha \tilde{B}$



$L(A) \cap L(B) \neq \phi$

$S \to aA / aB$

$A \to aA / \epsilon$

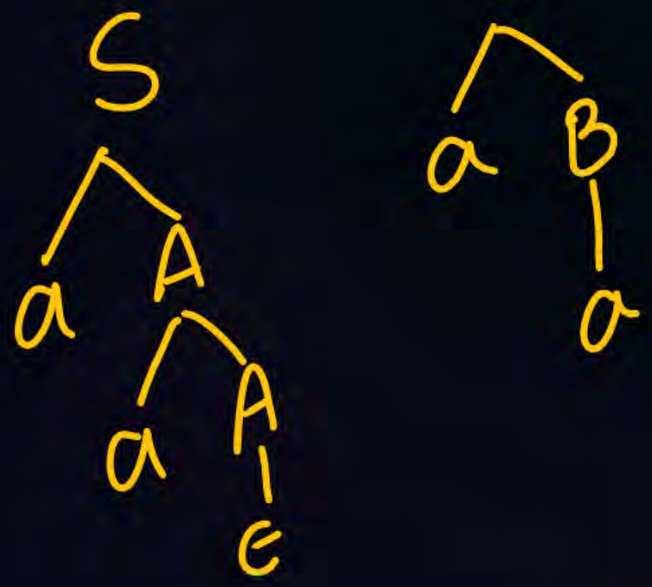
$B \to a$

$A \overset{*}{\Rightarrow} a^*$

$B \overset{*}{\Rightarrow} a$

$a^* \cap = \{a\}$

aa





$S \rightarrow aaA / aaB$ unamb?

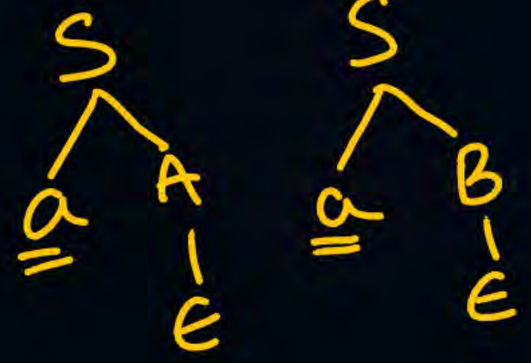
$A \rightarrow aA/a$

$B \rightarrow bB/b$

$A \overset{*}{\Rightarrow} a^+ \quad \cap = \phi$

$B \overset{*}{\Rightarrow} b^+$

Left factoring is not always ambiguous



$\epsilon \quad \epsilon$

$S \to aA/aB$

$A \to aA/\epsilon \quad A \overset{*}{\Rightarrow} a^* \quad n = \{\epsilon\}$

$B \to bB/\epsilon \quad B \overset{*}{\Rightarrow} b^*$

amb

$aa^* + ab^*$

$a^+ + ab^*$

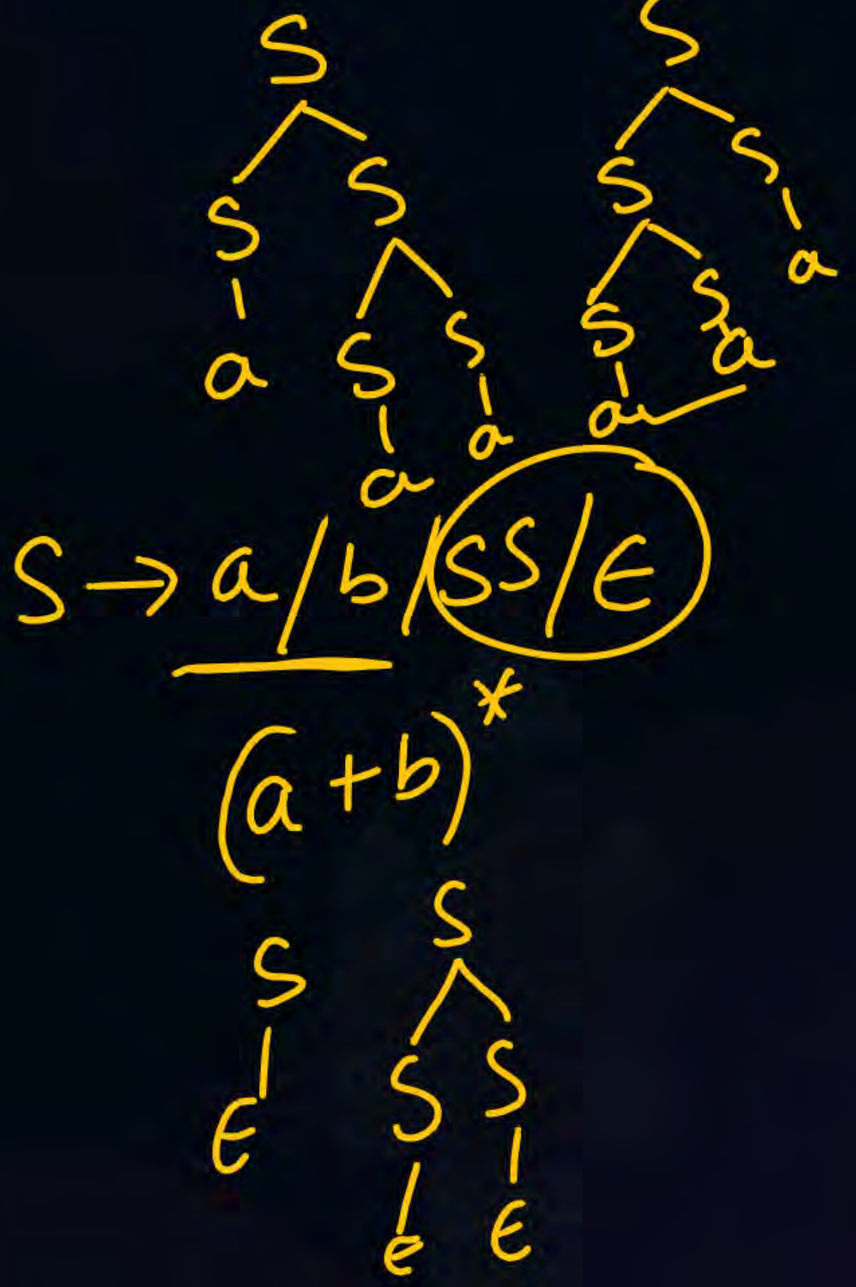
Rule 3:
Presence of SS

✓ $S \rightarrow ab/ba/SS/\epsilon$ ✓

$(ab+ba)^*$

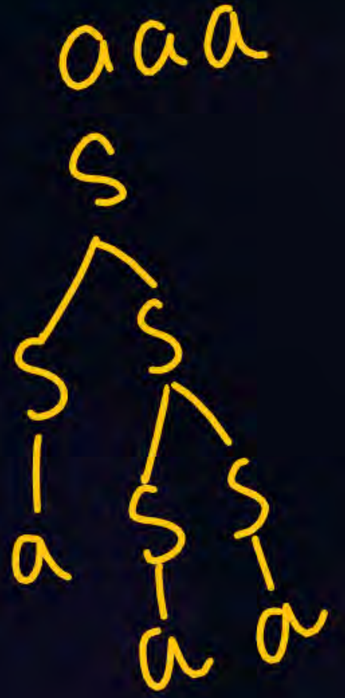
✓ S → ε

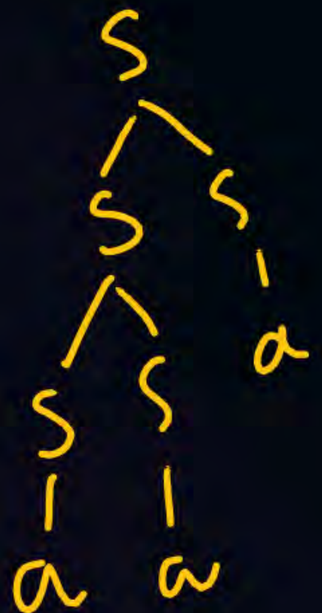
S → S S → ε ε



$S \rightarrow ab / ba / SS / a$

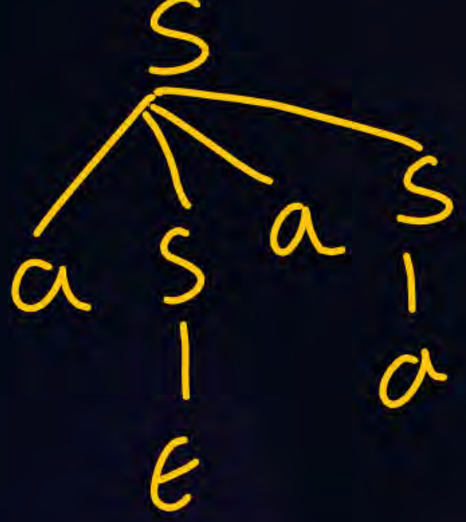
aaa

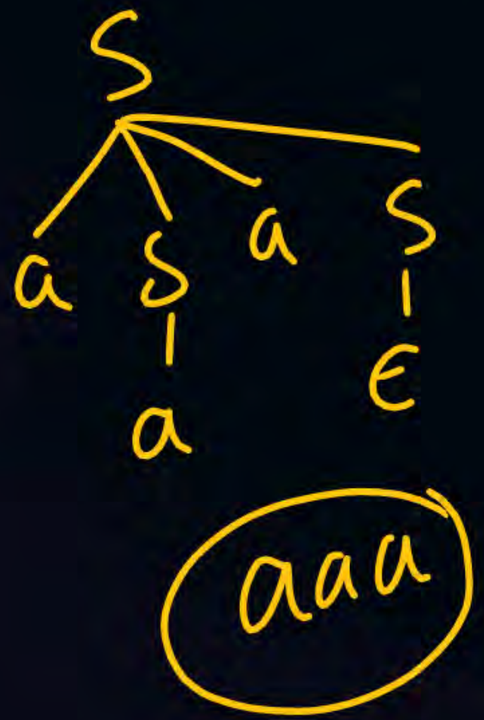


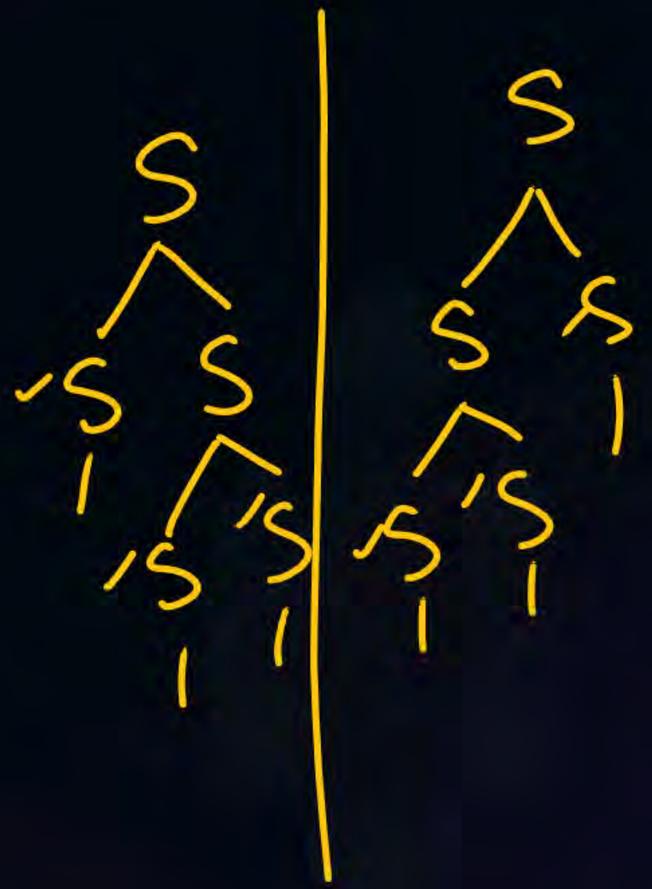


$S \rightarrow aSaS/a/\epsilon$

ambiguous

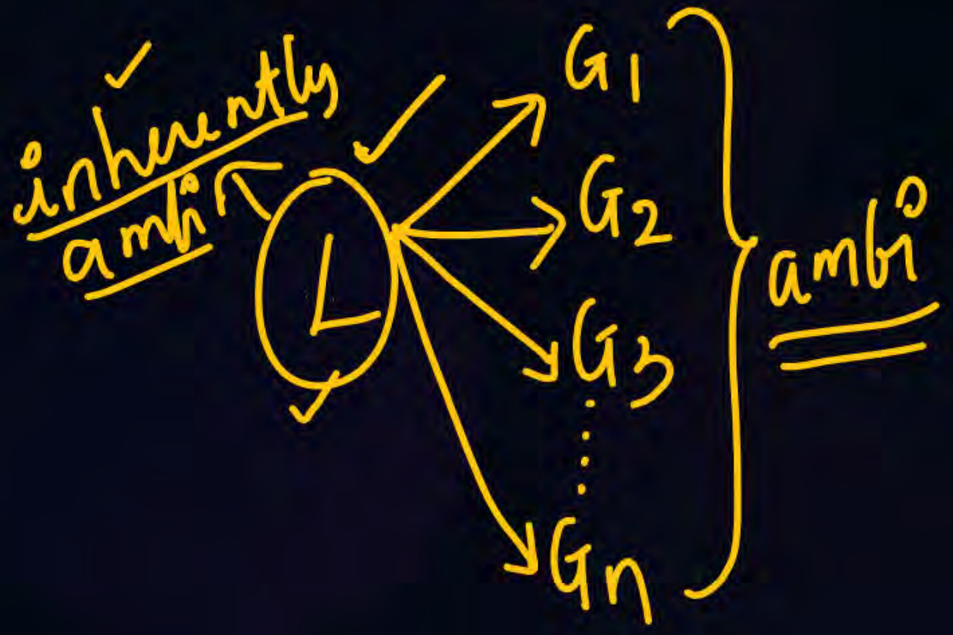






Grammar ambiguity → $\exists w$ → 2 ∄ mode DT

Language ambiguity



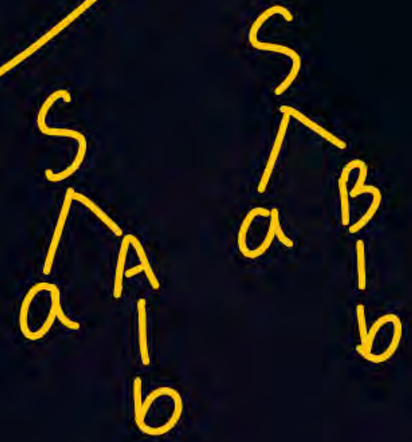


unambi

$L = \{ab\}$ ✓

$S \to aA / aB$
$B \to b$
$A \to b$ ✓

ambi grammar ✓



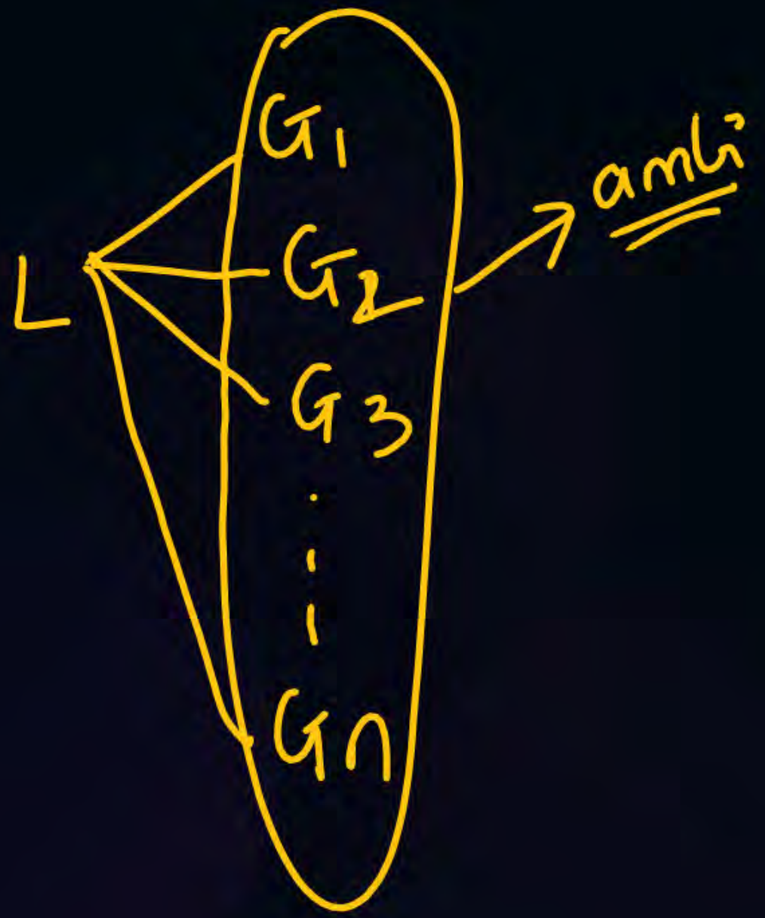
$S \to ab$ ✓

unambi gram for the same lang

If all the grammars of a language are ambiguous
then the language is ambiguous
Ambiguous languages are called inherently ambiguous

If a language is inherently ambiguous the all the grammars of the language are ambiguous

If ∃ atleast one unambiguous grammar for a language then the language is unambiguous


L
G1
G2
G3
Gn
amli

Rules to know if a language is ambiguous :-

$L = \{aa, bb, cc\}$

un $S \to aa/bb/cc$

Reg, DCFL



Rules to know if a language is ambiguous :-

1) L should be union of two DCFL languages

Ex : $(a^n b^n c^m) \cup (a^n b^m c^m)$

$L = L_1 \cup L_2$

2) $L_1 \cup L_2$. Should not collapse into a single language

$L_1 = \acute{L}_1 \cup \bar{L}_2$

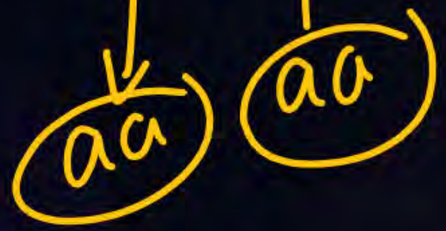
$a^n b^n \cup a^* b^* \longrightarrow$ collapse $a^* b^*$

3) $L_1 \cap L_2 \neq \phi$

$= \infty$

$L = L_1 \cup L_2$

$S \rightarrow S_1 / S_2$



$S_1 \overset{*}{\Rightarrow} L_1 \quad \cap = \phi$

$S_2 \overset{*}{\Rightarrow} L_2$

$L = L_1 \cup L_2$

$S_1 \overset{*}{\Rightarrow} L_1$

$S_2 \overset{*}{\Rightarrow} L_2$

$L_1 \cap L_2 = \infty$

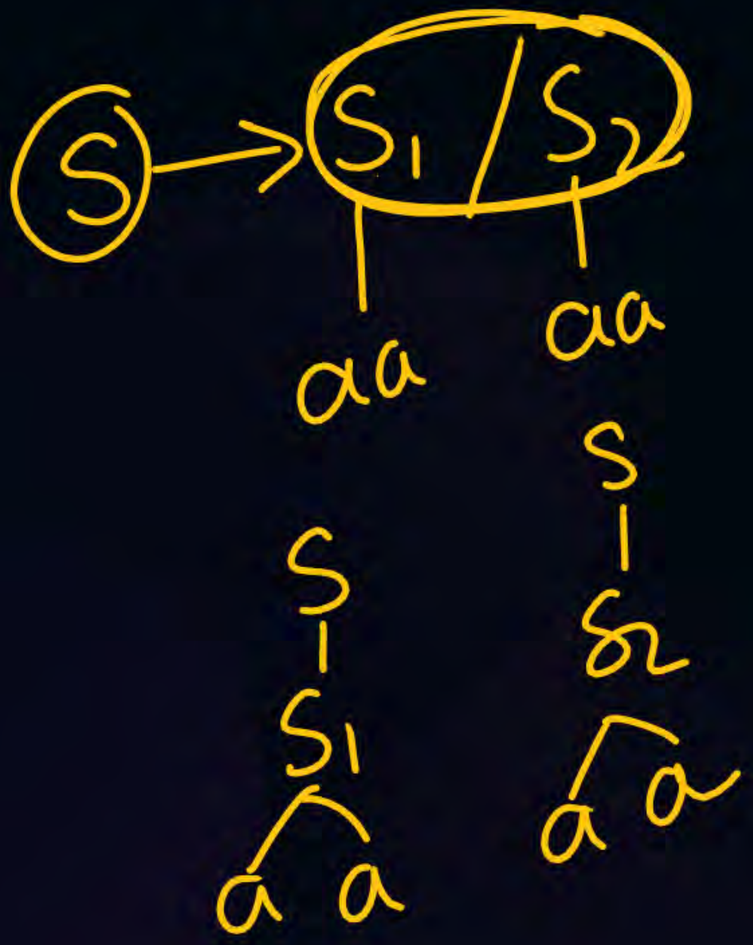
$S \rightarrow S_1 / S_2$

aa aa





$L = L_1 \cup L_2$ ✓



$S_1 \overset{*}{\Rightarrow} L_1$

$S_2 \overset{*}{\Rightarrow} L_2$

$L_1 \cap L_2 = \phi$ ✓

$L = a^n b^n \cup a^n b^{2n} / n \geq 0$ ✓

$a^n b^m$ / $m = n$ (b) $m = 2n$

1) $L = L_1 \cup L_2$

2) $L_1 \cup L_2$ × Collapse

X 3) $L_1 \cap L_2 \to \infty$ $\{\epsilon\}$ → finite

$S \to A / B$

$A \to aAb / \epsilon$

$B \to aBbb / \epsilon$

↓ ambi°

S I A I E

S I B I E

$L = a^n b^n \cup a^n b^{2n} / n \geq 0$ ✓

$a^n b^m$ / $m = n$ (b) $m = 2n$

1) $L = L_1 \cup L_2$

2) $L_1 \cup L_2$ × Collapse

~~3~~) $L_1 \cap L_2 \rightarrow \{\epsilon\}$ → finite

$S \rightarrow A / B / \epsilon$

$A \rightarrow aAb / ab$

$B \rightarrow aBbb / abb$

S I A E S I B E

↓ ambi°

$a^n b^n \cup a^{2n} b^{2n} / n \geq 0$

1) $L = L_1 \cup L_2$

2) $L_1 \cup L_2$ → should not collapse

$a^n b^n \cup a^{2n} b^{2n} = a^n b^n$

$a^{2n} b^{2n} \subseteq a^n b^n$

$a^n b^n = \{\epsilon, ab, aabb, aaabbb, \ldots\}$

$a^{2n} b^{2n} = \{\epsilon, aabb, aaaabbbb \ldots\}$

$S \rightarrow S_1 / S_2$

$S \rightarrow aSb / \epsilon$

↓

Unambiguous

$a^n b^n c^m \cup a^m b^n c^n / n, m \geq 0$

1) $L = L_1 \cup L_2$
2) X (Ambiguity)
3) $a^n b^n c^m \cap a^m b^n c^n = a^n b^n c^n$ = ∞

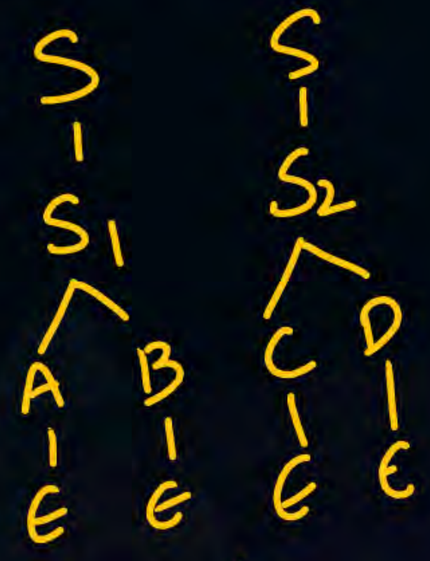
{abc ✓
aabbcc ✓
aaabbbccc ✓

$S \rightarrow S_1 / S_2$

$S_1 \rightarrow AB/\epsilon$
$A \rightarrow aAb/\epsilon$
$B \rightarrow cB/\epsilon$

$S_2 \rightarrow CD/\epsilon$
$C \rightarrow aC/\epsilon$
$D \rightarrow bDc/\epsilon$

→ Grammar ambiguous

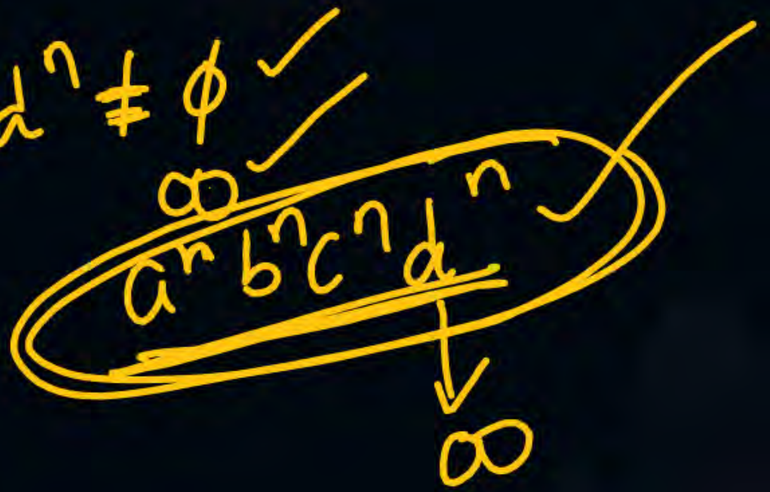


$a^n b^n c^m d^m \cup a^n b^m c^m d^n \; m,n \geq 0$

CSL

$L \neq \phi$ < Finite / Finite

1) $L = L_1 \cup L_2$
2) X closure property
3) $a^n b^n c^m d^m \cap a^n b^m c^m d^n \neq \phi$ ✓



$S \to X/Y$

$X \to AB$

$A \to aAb/\epsilon$

$B \to CBd/\epsilon$

$Y \to C$

$C \to aCd/D$

$D \to bDc/\epsilon$

$L \neq \phi$ < finite — unam
infinite — am

$L: a^n b^n c^n d^m \cup a^n b^m c^n d^m$ → CFL ✗

→ CSL ✓

$a^+ \cup b^+$

$S \to A/B$

$A \to aA/a$

$B \to bB/b$

1) $L = L_1 \cup L_2$

2) ✗ collapsing

3) $a^+ \cap b^+ = \phi$ → unambi

ϕ → unambi

$\cap$ ← ϕ – unambi

finite – unambi

∞ – ambiguous

→ Sometimes ambiguity is not a problem of the grammar, the underlying language may be ambiguous i.e the langage is inherently ambiguous.

→ If a language is unambiguous, there will at least one grammar for the langage that is unambiguous

$a^n b^m / n > m \cup a^n b^m / n < m$

1) $L = L_1 \cup L_2$

X2) Collapse

$a^n b^m / n \neq m$

unamb^{o} $a^+ a^n b^n / a^n b^n b^+$

$S \rightarrow S_1 / S_2$

$S_1 \rightarrow AB$

$A \rightarrow aA/a$

$B \rightarrow aBb/ab$

$S_2 \rightarrow BC$

$C \rightarrow bC/b$

→ A language has a grammar which is ambiguous then the language is ——

a) ambi b) unambi c) No Comment d) left recursive

→ A language has a grammar which is unambiguous then the languge is

a) ambi b) unambi c) no Comment d) None

If a grammar G is ambiguous and language L is ambiguous
Can we Convert G into unambiguous grammar?

T/F

NO

$a^m b^n / m > n \cup a^m b^n / m < n \cup a^m b^n / m = n$

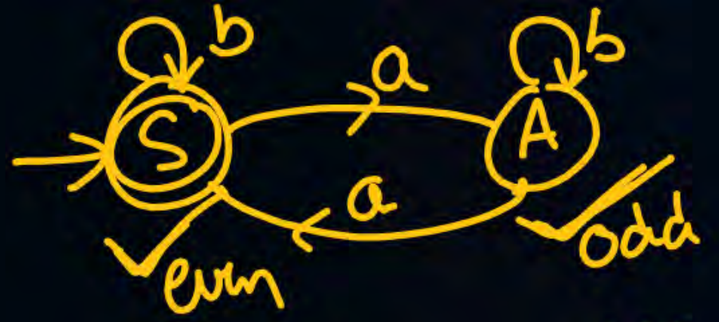
collapse

$a^* b^* \longrightarrow$ Reg $\rightarrow$ unambi°

5 min break

$S \to bS/aA/\epsilon$

$A \to aS/bA$



The myhill-Nerode equivalence classes generated

a) $\{w \in (a+b)^* / \#_a(w) \text{ is even}\}$ and $\{w \in (a+b)^* / \#_a(w) \text{ is odd}\}$

b) $\{w \in (a+b)^* / \#_b(w) \text{ is even}\}$ and $\{w \in (a+b)^* / \#_b(w) \text{ is odd}\}$

c) $\{w \in (a+b)^* / \#_a(w) = \#_b(w)\}$ and $\{w \in (a+b)^* / \#_a(w) \neq \#_b(w)\}$

d) $\{\epsilon\}, \{wa / w \in (a+b)^*\}$ and $\{wb / w \in (a+b)^*\}$

Let L be a regular language. Consider the following

I) $Repeat(L) = \{ww / w \in L\}$

II) $Prefix(L) = \{u / \exists v : uv \in L\}$

III) $Suffix(L) = \{v / \exists u : uv \in L\}$

IV) $half(L) = \{u / v : |\exists v| = |u| \text{ and } uv \in L\}$

Which of following is non reg

a) Both I and IV

b) only I

c) only IV

d) Both II and III

Consider the Reg language $L=(111+11111)^*$ The min num of states in DFA

a) 3 b) 5 c) 8 d) 9

Which of the following is true

X a) Every Subset of a regular set is regular

✓ b) Every finite Subset of a non regular set is regular

c) union of two non regular sets is not regular

d) Infinite union of finite sets is regular

$a^n b^n \subseteq (a+b)^*$ ✓

$(ab), (aabb), (aaabbb), \ldots : a^n b^n$

$a^n b^n \cup a^n b^n = (a+b)^*$

$a^n b^n$: CFL

$a^n b^n$: NR

CFL

NR

A minimum state DFA accepting the language $L = \{w / w \in (0,1)^*, \#_0$ and $\#_1$ in w are divisible by 3 and 5 respectively has

a) 15 states b) 11 states

c) 10 states d) 9 states

Which of the following languages are regular?

a) $\{ww^R / w \in (0,1)^+\}$

b) $\{ww^Rx / x,w \in (0,1)^+\}$

c) $\{wxw^R / x,w \in (0,1)^+\}$

d) $\{xwx / x,w \in (0,1)^+\}$

$S \to xB / yA$

$A \to x / xS / yAA$

$B \to y / yS / xBB$

$S \to aA / bB$

Which strings are generated

a) xxyy b) xxxy

c) xyx d) yxx

THANK - YOU